CUSTOMER RETENTION & CHURN ANALYSIS

Task 2 | Data Science & Analytics | Future Interns

June 2026

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Domain	Data Science & Analytics	**Task**	Task 2
Tools	Python, Matplotlib, Seaborn, Plotly	**Date**	June 2026

1. Project Overview

This report presents a comprehensive Customer Retention and Churn Analysis on a telecom subscription dataset as part of Future Interns Task 2. The objective is to identify churn patterns, retention drivers, and provide actionable recommendations to reduce customer loss and protect revenue.

Objectives:

- Analyze churn patterns across customer segments
- Identify high-risk profiles and key retention drivers
- Perform cohort-based retention trend analysis
- Quantify revenue at risk from churned customers
- Deliver data-backed business recommendations

2. Key Performance Indicators

Metric	Value	Metric	Value
Total Customers	1,500	**Avg Tenure (All)**	35.9 months
Active Customers	1,023	**Avg Tenure (Churned)**	31.3 months
Churned Customers	477	**Avg Tenure (Active)**	38.0 months
Overall Churn Rate	**31.8%**	**Avg Monthly Charge**	Rs. 64.53
Revenue at Risk	**Rs. 1,053,679**	**Total Features**	18

3. Churn Analysis by Key Segments

3.1 By Contract Type

Contract type is the strongest churn predictor. Month-to-Month customers are significantly more likely to churn compared to annual contract holders.

Contract Type	Churn Rate (%)	Risk Level
Month-to-Month	39.9%	**HIGH**
One Year	24.2%	**MEDIUM**
Two Year	23.0%	**LOW**

3.2 By Internet Service

Internet Service	Churn Rate (%)	Observation
DSL	26.7%	Moderate
Fiber Optic	38.5%	Highest - Quality Issue
No Internet	25.4%	Lowest

3.3 By Payment Method

Payment Method	Churn Rate (%)
Electronic Check	40.3%
Bank Transfer	27.6%
Credit Card	27.1%
Mailed Check	27.0%

3.4 By Tenure Group

New customers are at highest risk. Churn drops significantly after the first year and becomes very low beyond 36 months.

Tenure Group	Churn Rate (%)	Retention Signal
0-6 months	52.9%	Critical Risk
7-12 months	40.7%	High Risk
13-24 months	30.9%	Moderate
25-36 months	35.9%	Stable
36+ months	25.1%	Very Loyal

4. Churn Reasons Analysis

Among 477 churned customers, price sensitivity and competitor attraction are the dominant exit reasons.

Rank	Churn Reason	Customers	Share (%)
1	Financial Constraints	95	19.9%
2	Moved Away	85	17.8%
3	Switched to Competitor	84	17.6%
4	Better Price Elsewhere	83	17.4%
5	Lack of Features	69	14.5%
6	Poor Customer Service	61	12.8%

5. Analytics Dashboard

The dashboard below presents all key churn metrics and trends in a single view — covering churn distribution, segment-wise rates, cohort retention trend, scatter analysis, and regional performance.

6. Key Insights

- Contract Type Drives Churn: Month-to-Month customers churn at 39.9% vs only 23.0% for Two Year contracts — a 5x difference.
- Early Tenure is the Danger Zone: Customers in first 6 months churn at ~52.9%. After 36 months this drops to ~25.1%.
- Fiber Optic Quality Issue: Despite being premium, Fiber Optic churns at 38.5% vs DSL 26.7%.
- Payment Method = Commitment Level: Electronic Check users churn at 40.3%. Auto-pay users are significantly more loyal.
- Multi-Product Customers Are Loyal: Customers with 3+ products churn far less — bundling reduces churn.
- Revenue at Risk: Rs. 1,053,679 in total charges is at risk from churned customers.

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7. Business Recommendations

Priority	Recommendation	Expected Impact
HIGH	Offer discounts to convert Month-to-Month customers to annual contracts	Reduce churn ~15%
HIGH	Launch structured 90-day onboarding program for new customers	Reduce early churn ~20%
MEDIUM	Conduct Fiber Optic quality audit and implement improvements	Reduce fiber churn ~10%
MEDIUM	Incentivize Electronic Check users to switch to auto-pay	Reduce payment churn ~8%
LOW	Cross-sell product bundles to single-product customers	Increase product adoption
LOW	Introduce loyalty rewards at the 24-month customer milestone	Improve retention

8. Conclusion

This analysis of 1,500 telecom customers reveals a 31.8% overall churn rate with Rs. 1,053,679 in revenue at risk.

Contract type, early tenure, and service quality are the primary churn drivers. Targeted interventions — converting Month-to-Month customers to longer contracts and strengthening the onboarding experience — can significantly reduce churn and improve customer lifetime value.